

1:Error detection and correction:catch the corrupted message

AIM

To detect transmission errors using parity bits in python

PROGRAM

```
import random
#convert text to binary
def text_to_binary(text):
    return "".join(format(ord(i),'08b')for i in text)

#Even parity
def parity_bit(binary):
    ones = binary.count('1')
    return '0' if ones % 2 == 0
    else '1'
#flip one random bit
def flip_bit(binary):
    Index = random.randint(0,len(binary)-1)
    Bit = '1' if binary[Index]!='0'
    Else '0'
    New_binary = binary[:Index] + bit + binary[Index+:]
    Return new_binary, index
#verify parity
def check_parity(binary,parity):
    total = binary.count('1')
    received = total+ int(parity)
    if received % 2 == 0:
        return "OK"
    Else:
        Return "ERROR DETECTED"
#Checksum
def checksum(text):
    return sum(ord(c)for c in text)
#Main
message = input("Enter message:")

binary = text_to_binary(message)
parity = parity_bit(binary)

print ("Original Binary :", binary)
print("parity Bit:", parity)

Received , pos = flip_bit(binary)

print("\nNoise added")
print("bit flipped at position:", pos)

print("Received Binary:", received )

print("parity check:",check_parity(received, parity))

print("\nChecksum:",checksum(message))
```

OUTPUT:

Enter message: HELLO

Original binary:

0100100001000101010011000100110001001111

Parity bit:

0

Noise added bit flipped at position: 18

Received binary:

0100100001000101000011000100110001001111

Parity check:

ERROR DETECTED

Checksum:

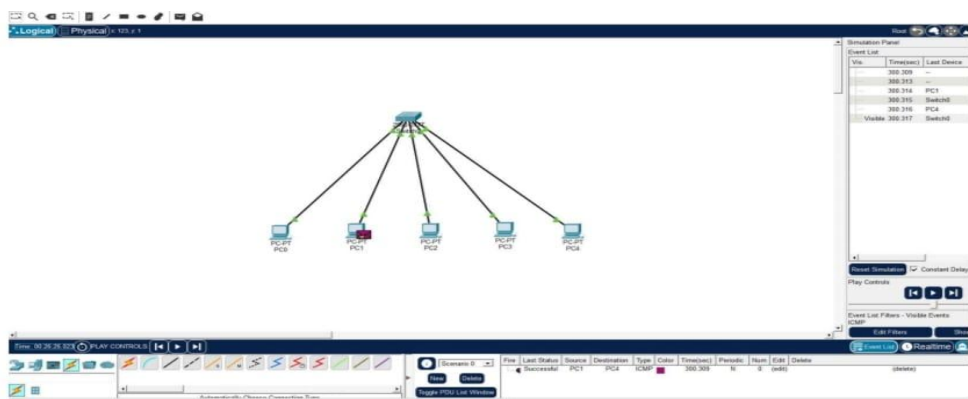
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OBSERVATION:

The transmitted messages were intentionally corrupted by flipped one random bit. Both the parity bit and checksum detected the errors successfully. The experiment shows that checksum is generally more reliable than parity for detecting transmission errors.

2:LAN & multiple access-”set up and test a mini home/lab network”:

This is topology:



This is IP configuration:

This is ping test

```
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Observation:

During the experiment ,all five PCs connected through the switch communicated successfully using the ping.the switch forwarded data only to the destination PCs,reducing unnecessarybtraffic.when the switch was replaced with a hub,data was broadcast to all connected PCs,increasing network traffic and the change of collisions.this show the switches provide better performance and efficiency than hubs.therefore,switches are preferred in modern LANs.

3:WIRELESS BASIC “survey your own wi-fi world”

Sample wi-fi comparison table

SSID	Signal Strength	Channel	Wi-Fi Standard
Home_WiFi	-40 dBm	36	Wi-Fi 6 (802.11ax)
Airtel_Xstream	-55 dBm	6	Wi-Fi 5 (802.11ac)
JioFiber	-60 dBm	11	Wi-Fi 5 (802.11ac)
Hostel_WiFi	-70 dBm	1	Wi-Fi 4 (802.11n)
TP-Link	-75 dBm	44	Wi-Fi 6 (802.11ax)

OBSERVATION:

The wi-fi scan showed that nearby network have different signal strengths and wi-fi standards.networks with stronger signals (closer to -40 dBm) provide better performance than weaker signals(around -70 dBm).wi-fi 6 networks offer higher speed,lower delay,and better performance in crowded areas compared to older wi-fi standards.